ADVANCED PREDICTIVE MODELLING FOR CUSTOMER CHURN IN BANKING: A COMPARATIVE ANALYSIS OF MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

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Research Proposal

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ABSTRACT

Customer churn in banking is a costly problem, as retaining existing customers is significantly more affordable than new customer acquisition. With the help of predictive analytics algorithms, banks can easily identify at-risk customer at an early stage and intervene with appropriate retention planning. Using a public Kaggle dataset of over 5,000 banking customers, this paper studies churn prediction applying both advanced machine learning (ML) and deep learning (DL) techniques.

The methodology combines data preprocessing, model training with appropriate hyperparameter tuning, model evaluation and interpretability analysis. Data preprocessing covers missing value imputation, one hot encoding for low-cardinality categorical feature, feature scaling and class imbalance treatment using SMOTE and GAN-based method. While Logistic Regression acts as baseline, Decision Tree, Random Forest, and XGBoost are candidates for ML, Artificial Neural Networks as baseline and Deep Neural Networks are selected for DL. Hyperparameter tuning is applied with Optuna, and model performance is evaluated using accuracy, precision, recall, F1-score, AUC, and execution time.

To ensure actionable insight, the final model must be interpreted and understood correctly. It is necessary to employ interpretability analysis such as SHAP. This enables both global and local explanations of model predictions. The paper results are expected to provide a fair view of performance versus computational efficiency, guiding practitioners in figuring out the best predictive framework for real-time deployment in banking. It is noted, however, that deep learning models are not subjected to full hyperparameter optimisation due to dataset size and computational constraints, representing a conscious trade-off within the study's scope.

1. BACKGROUND

In banking sector, customer churn, the behavior by which clients terminate their relationship with a bank, remains one of the most pressing strategic challenges. New customer acquisition cost is approximately about five and seven times more costly than old customer retention cost thus even small retention increase can result in substantial bottom-line improvements result in substantial bottom-line improvements (Bhuria *et al*., 2025; Singh, H., *et al*., 2024). Churn has several impacts. Besides its direct impact on revenue from deposit, lending and fee-based services, it has two additional indirect losses. It is a long-term loss because the loss of cross sales opportunities. The other is the loss of brand loyalty (AbdelAziz *et al*., 2025; Abbas *et al*., 2023).

Nowadays, due to the rise of fintech rivals, neo bank, and technology-driven service models, the competitive intensity in retail and digital banking has increased sharply. Especially, with the technology innovation, mobile banking or digital banking enables low barrier to entry, allowing customers to compare and move to alternative banks in real time (Murindanyi *et al*., 2025). Therefore, high value customer retention is no longer a simple marketing function, but it requires a cooperation across functions. The recent customer retention strategy usually includes analytics, operations and strategic planning (Advances in Economics, Management & Political Sciences, 2024).

While banking has its own structural drivers of churn, similar experiences are observed in other industries. In telecommunications, because of high competition and low switching barriers, various churn prediction models ranging from logistic regression to advanced ensemble methods have been comprehensively developed and demonstrated strong performance (Boozary, 2025). Similarly, churn prediction models, based on purchase frequency, browsing behaviour, and promotional response patterns, are also well implemented in e-commerce to detect high risk of platform leaving. Those models that are trained using gradient boosting and deep neural networks algorithms, have shown particularly strong recall (Ahmed *et al.*, 2023).

There is clear proof saying that despite the domain differences, the methodological challenges such as class imbalance, feature selection, and the trade-off between interpretability and accuracy are remarkably consistent. These shared analytical challenges have powered the search for more sophisticated and generalisable predictive frameworks that can be applied across multiple industries, including banking.

Predictive analytics has become an essential competence for modern retention planning in other industries. At early stage, traditional statistical models such as logistic regression were used for churn prediction in banking. They were good for their interpretability yet their ability in capturing complex non-linear behaviors were very limited (AbdelAziz *et al*., 2025; Patel, S., *et al*., 2024). Thanks to the rapid evolution of machine learning (ML), ensemble methods such as random forests and gradient boosting has been introduced. They leverage multiple base learners to improve accuracy and generalization (Singh, P., *et al*., 2025). These models have demonstrated competitive performance in banking datasets, as well as in related sectors like telecommunications and e-commerce (Boozary, 2025). Nevertheless, when feature relations are highly complex or when predictive signals are embedded in high-dimensional, noisy, or heterogeneous data a limitation, even the most advanced ML ensembles can struggle. Hence, deep learning has emerged as a powerful approach.

More recently, deep learning (DL) approaches have expanded the analytical capabilities by enabling the automatic extraction of complex feature interactions from high-dimensional data. Architectures such as artificial neural networks (ANNs) and deep neural networks (DNNs) have achieved notable improvements in recall and F1-score, especially when combined with appropriate feature scaling and optimization (Domingos, Ojeme & Daramola, 2021). Hybrid models that integrate convolutional neural networks (CNNs) with recurrent structures like bidirectional long short-term memory (BiLSTM) have further demonstrated strong performance in identifying subtle churn signals (Ahmed *et al*., 2023; Zhang *et al*., 2024). In comparative experiments, Singh *et al*., (2023) and Adamu *et al*., (2025) report that DL can outperform gradient boosting methods under certain configurations, although at a higher computational cost.

Although DL performance is impressive, practical application in banking is considerably low because of two main reasons. DL models usually consume significant computational resources and longer training times and are inherently less interpretable than traditional ML (Basit *et al*., 2024). Especially, this interpretability challenge is getting more significant in regulated environments, where explainable AI (XAI) is becoming a compliance expectation (Tékouabou, 2022; Ma *et al*., 2022). By integrating interpretability techniques such as SHAP values into churn models few papers such as Li, X., *et al*. (2025) and Singh, A., *et al*. (2024) address this gap between predictive accuracy and regulatory transparency.

Class imbalance, where churn customers play only a small proportion of the total customer base, is another repeated challenge in churn prediction. During training, models may be biased toward the majority class, reducing the ability to detect true churners, even when overall accuracy appears high. A range of techniques have been proposed to address this, including synthetic minority over-sampling (SMOTE), adaptive synthetic sampling (ADASYN), and more recently, generative adversarial network (GAN)-based oversampling (Adiputra, Wanchai & Lin, 2025). In addition, cost-sensitive learning approaches, where misclassifications of churners incur higher penalties, have been shown to improve recall and business impact in ensemble models (Bhuria *et al*., 2025; Abubakar *et al*., 2024).

Hyperparameter tuning plays a critical role in maximizing the potential of both ML and DL models. Although traditional methods such as grid search and random search are still common, they may be inefficient for complex models with large parameter spaces. Optuna, a more advanced framework, has been introduced to restructure the search process, applying pruning strategies to terminate underperforming trials early (Domingos, Ojeme & Daramola, 2021; Patel *et al*., 2023). As highlighted by Marmion *et al*. (2012) and Molnar (2020), model interpretability is considered together with hyperparameter tuning strategies to ensure that tuning does not compromise transparency.

Despite extensive prior work, there are few gaps remain. First, although both ML and DL have been used to churn prediction in banking, there is a lack of direct head-to-head comparisons conducted under identical experimental setups (AbdelAziz *et al*., 2025; Basit *et al*., 2024). Second, execution time is usually underestimated and so frequently overlooked as a performance metric regardless its significance for real-time inference systems (Singh, P., *et al*., 2025). Third, there is only a limited empirical evidence evaluating the effect of preprocessing, imbalance handling, and hyperparameter optimization combination on model performance in banking sector (Adiputra, Wanchai & Lin, 2025; Patel *et al*., 2023).

By combining three actions at once, this thesis seeks to narrow these gaps. First, it evaluates both ML and DL algorithms on a public banking churn dataset. Second, it applies sophisticated sampling approaches and rigorous hyperparameter tuning. Lastly, the paper also comprehensive evaluates predictive performance and operational efficiency. The first goal of this paper is to assists practitioners in model selection that balance performance, interpretability, and deployment feasibility. Another goal is to contribute to an academic understanding of predictive modeling in financial services.

2. LITERATURE REVIEW

Customer churn prediction is an interesting problem that has always attracted many researcher from all over the world and across all industries. Each study brings domain specific insight and methodological innovations. In banking, this problem has grown from traditional statistical approaches to advanced machine learning (ML) and deep learning (DL) architectures, resulting from a bigger trends from other sectors such as telecommunications and ecommerce (Boozary, 2025; Ahmed *et al*., 2023). As stated in the background sections, these industries share common analytical challenges class imbalance, the need for robust feature selection, and the trade-off between interpretability and predictive accuracy despite differences in customer behaviour and operational contexts. As supported by the cross-industry evidence, it is reasonable for evaluating a various set of ML and DL approaches in the banking sector. Besides the motivation to benchmark performance, it is also to access their adaptability to the sector's unique regulatory and customer engagement requirements.

In banking, predictive analytics for churn must address the dual challenge of delivering high predictive accuracy while meeting operational, regulatory, and interpretability requirements. This section synthesizes prior work into three main themes: machine learning approach, deep learning approach, performance metrics and imbalance handling. The table below describe a high-level summary of prior literatures.

Table 1: Literature review summary

Section	Problem	Methodology	Dataset Used	Interpretation of Results	Citation
Machine Learning Approaches	Low accuracy and overfitting in simple decision trees	Logistic Regression, Decision Trees	Banking dataset	Logistic regression provides interpretability but lower accuracy than ensembles	Abbas *et al*. (2023)
	Benchmarking ML vs DL for churn prediction	Gradient Boosting, Random Forest	Banking dataset	Gradient boosting outperforms bagging ensembles when tuned	AbdelAziz *et al*. (2025)
	Evaluating baseline ML classifiers	Logistic Regression, Decision Trees	Banking dataset	Logistic regression valued for transparency, DTs for interpretability	Patel, S., *et al*. (2024)
	Improving prediction stability	Random Forest, XGBoost	Banking dataset	Ensembles improve generalisation and reduce overfitting	Singh, P., *et al*. (2025)
	Cross-industry churn application	Random Forest, boosting	Ecom dataset	Boosting techniques yielded highest accuracy	Boozary (2025)

	Cost-sensitive churn classification	SVM, Logistic Regression	Banking dataset	SVM captures nonlinear decision boundaries	Abubakar *et al*. (2024)
	Ensemble stability in churn models	Hybrid Ensembles (Voting, Stacking)	Telco dataset	Hybrid ensembles yield more stable predictions	Bhuria *et al*. (2025)
	Multi-sector churn analysis	Random Forest, ensembles	Banking & Telecom dataset	Ensemble methods robust across domains	Murindanyi *et al*. (2025)
Deep Learning Approaches	Manual feature engineering limits complexity	ANN	Banking dataset	ANN improves recall over ML baselines	AbdelAziz *et al*. (2025)
	Exploring DL model effectiveness	DNN	Banking dataset	DNN achieves higher F1 when tuned properly	Adamu *et al*. (2025)
	Comparative DL for churn prediction	DNN	Banking dataset	DNN captures higher-order feature interactions	Singh, H., *et al*. (2024)
	Feature-rich DL for churn	DNN	Banking dataset	DNN shows significant accuracy gains	Basit *et al*. (2024)
	Sequential dependencies in churn	CNN–BiLSTM hybrid	Telco dataset	CNN–BiLSTM improves recall by capturing temporal dependencies	Ahmed *et al*. (2023)
	Hybrid network for churn	CCP-Net	Banking dataset	CCP-Net detects spatial–temporal churn patterns	Zhang *et al*. (2024)
	Temporal patterns in churn	LSTM	Telco dataset	LSTM improves sequence modelling in churn	Sezer *et al*. (2018)
	Complexity and interpretability	DL with feature selection	Banking dataset	Feature selection reduces complexity and cost	Ma *et al*. (2022)
Performance Metrics	High cost of misclassification	Accuracy, Recall, F1-score	Banking dataset	Recall is most critical for business impact	Bhuria *et al*. (2025)
	Benchmarking metrics in churn	Accuracy, Precision, Recall	Banking dataset	Precision–recall trade-off must be balanced	AbdelAziz *et al*. (2025)

	Emphasis on churn detection	Recall, F1-score	Banking dataset	Recall improvements strongly affect business outcomes	Singh, P., *et al*. (2025)
	Metric evaluation across domains	Accuracy, Recall, F1-score	Banking + Telecom dataset	F1 balances recall–precision better than accuracy	Adiputra, Wanchai & Lin (2025)
Imbalance Handling	Severe class imbalance	SMOTE oversampling	Banking dataset	Improves recall by increasing minority samples	Tékouabou (2022)
	Advanced imbalance correction	GAN-based oversampling	Banking dataset	GAN generates realistic minority samples	Ali, M., *et al*. (2024)
	Data imbalance in churn	Cost-sensitive learning	Telco dataset	Penalising churner misclassification improves recall	Imran *et al*. (2023)
	Hybrid ensemble imbalance correction	RUSBoost	Banking dataset	Combines under-sampling + boosting effectively	Bhuria *et al*. (2025)
ML Explainability	Lack of interpretability in black-box models	SHAP, interpretable ensembles	Banking dataset	SHAP improves transparency with minimal accuracy loss	Li, X., *et al*. (2025)
	Interpretable churn prediction	LIME + ML models	Banking dataset	LIME explanations enhance trust in churn predictions	Singh, A., *et al*. (2024)
	Oversampling + interpretability	SMOTE with interpretable ensembles	Banking dataset	Improves recall while preserving interpretability	Tékouabou (2022)
Evaluation Strategy	Over-optimistic evaluation	Stratified k-fold CV	Banking dataset	Cross-validation provides robust estimates	Ahmed *et al*. (2023)
	Statistical confirmation of models	Significance testing	Banking dataset	Validates differences between models	Marmion *et al*. (2012)
	Interpretability as complement	Explainability frameworks	Banking dataset	Interpretability enhances trust in high-performing models	Molnar (2020)

2.1. Machine Learning (ML) Approaches for Churn Prediction

For churn prediction problem, it is nature for data to be well structure in tabular format, and ML algorithms are well known for their strong performance on that kind of data. Therefore, ML has been always a wise choice for churn prediction. Each ML algorithms are well known for their characteristics.

Logistic Regression (LR) is very good for its transparency and statistically simple (Abbas *et al*., 2023; AbdelAziz *et al*., 2025) and Support Vector Machines (SVM) is worthy for their ability to handle nonlinear decision boundaries (Abubakar *et al*., 2024). Decision Tree (DT) algorithm is the best for their interpretability and rapid inference, which make it a good choice for customer retention system deployment (Bhuria *et al*., 2025; Patel, S., *et al*., 2024). Those three models traditional, well-known ML algorithms, and they are acting as bassline. However, DT usually falls behind compared to ensemble methods (Singh, P., *et al*., 2025).

Random forest (RF) models, which combine multiple DTs through bagging and random feature selection, have been shown to improve accuracy and generalization while mitigating overfitting (Boozary, 2025). Gradient boosting algorithms, including Gradient Boosted Decision Trees (GBDT) and XGBoost, offer further refinements by sequentially refining weak learners (Adiputra, Wanchai & Lin, 2025; AbdelAziz *et al*., 2025; Li, X., *et al*., 2025). According to Patel *et al.,* (2023) if boosting methods are properly tuned, they may outperform bagging ensembles. Hybrid ensemble strategies such as voting classifiers and stacking have been applied to combine strengths from multiple algorithms, often yielding more stable predictions than any single model (Murindanyi *et al*., 2025).

In practice, one of the barriers to apply ML is the difficulty to understand them. Therefore, nowadays explainability has also begun as a focus in ML-based churn models. By integrating interpretable machine learning frameworks e.g. SHAP and LIME, into banking churn prediction, Li, X., *et al*. (2025) and Singh, A., *et al*. (2024) successfully enable transparency without significantly sacrificing accuracy. Tékouabou (2022) proposes an approach that combines SMOTE oversampling with interpretable ensemble models to improve both performance and explainability.

2.2. Deep Learning (DL) Approaches for Churn Prediction

Deep learning (DL) offers the ability to automatically learn complex feature interactions, reducing the need for manual feature engineering. Artificial neural networks (ANNs) have been applied in banking churn prediction with encouraging results, often outperforming baseline ML models in recall and F1-score (AbdelAziz *et al*., 2025; Domingos, Ojeme & Daramola, 2021; Adamu *et al*., 2025). Deep neural networks (DNNs), which add multiple hidden layers to ANNs, further enhance the modelling of high-order feature interactions (Singh, H., *et al*., 2024; Basit *et al*., 2024).

Hybrid DL architectures combine different network types to capture both spatial and temporal relationships in customer behavior. Ahmed *et al*. (2023) propose a CNN–BiLSTM hybrid for telco churn prediction, achieving substantial recall improvements, while Zhang *et al*. (2024) present CCP-Net, a hybrid neural network framework adaptable to banking datasets. Sezer *et al*. (2018) demonstrate that LSTM-based architectures can capture sequential dependencies, an approach relevant for modelling transaction histories in banking.

Recent studies are also exploring feature selection within DL pipelines to reduce model complexity and improve training efficiency. Abbas *et al*. (2023) demonstrate the successful adaptation of DL models to banking data through domain-specific preprocessing.

While DL models deliver strong predictive performance, they present challenges in interpretability and computational demands (Adiputra, Wanchai & Lin, 2025; Murindanyi *et al*., 2025). Ma *et al*. (2022) stress the importance of interpretable AI in banking, noting that DL models require post-hoc explanation methods to meet regulatory requirements.

2.3. Imbalance Handling and Performance Metrics

Class imbalance is a persistent problem in churn prediction across industries. There are two approaches, either data level or algorithm level. Data-level techniques such as SMOTE (Adiputra, Wanchai & Lin, 2025; Tékouabou, 2022) and GAN-based oversampling (Adiputra, Wanchai & Lin, 2025) have been used to increase minority class representation. Algorithm-level methods, including cost-sensitive learning (Bhuria *et al*., 2025) and probability calibration techniques (Abubakar *et al*., 2024), have been employed to prioritize churner detection.

The choice of performance metrics is critical in churn prediction, where misclassifying a churner carries a greater business cost than a false positive. There are four widely used metrics, which are accuracy, precision, recall and F1-score (Adiputra, Wanchai & Lin, 2025; Bhuria *et al*., 2025). In banking churn prediction, among those four metrics, recall is exclusively key. Because retaining an at-risk customer can return higher long-term value (AbdelAziz *et al*., 2025; Singh, P., *et al*., 2025).

Evaluation strategies normally involve k-fold cross-validation to ensure robust performance estimates. Ahmed *et al*. (2023) and Marmion *et al*. (2012) support for statistical significance testing to validate observed differences between models, while Molnar (2020) emphasizes interpretability as an essential complement to predictive metrics.

2.4. Research Gaps and Opportunities

After studying many papers, there are still some gaps needed to be deeply investigated and resolved. First, it is quite rare to find comparative studies, which evaluate ML and DL models under identical banking domain conditions (AbdelAziz *et al*., 2025; Basit *et al*., 2024).

Second, although execution time is an operationally critical factor for real-time interventions, it is usually ignored from and evaluation always focuses on predictive performance (Singh, P., *et al*., 2025).

Third, preprocessing, imbalance handling, and hyperparameter tuning are often applied in isolation rather than systematically combined and evaluated for their cumulative effect on performance (Adiputra, Wanchai & Lin, 2025; Patel *et al*., 2023). The adoption of advanced optimization frameworks such as Optuna remains limited in banking churn research, despite evidence of its efficiency (Domingos, Ojeme & Daramola, 2021).

Finally, besides predictive performance and execution time, the result explainability is also very important and should be taken with enough care. Studies like Li, X., *et al*. (2025), Singh, A., *et al*. (2024), and Ma *et al*. (2022) point to the need for churn prediction systems that are both accurate and interpretable, aligning with regulatory requirements and enhancing stakeholder trust.

3. RESEARCH QUESTIONS

This research is guided by four central questions derived from both literature gaps and the operational needs of banking institutions.

First, it presents a head-to-head comparison of how selected machine learning and deep learning models perform in churn prediction in banking sector under a reliable experimental basis. Despite there are few papers evaluate these models individually (AbdelAziz *et al*., 2025; Singh, H., *et al*., 2024; Adamu *et al*., 2025), few offer direct comparative analysis within the same data and processing conditions.

Second, the essential requirement of go-to-market churn prediction in real time retention system in banking sector is computational execution time must be quick. Although execution time is crucial factor, it is usually underestimated compared to predictive performance according to Singh, P., *et al*. (2025). This paper investigates the trade-off between those two factors and to re-evaluate the role of ensemble methods.

Third, it studies how class imbalance handling approaches and feature engineering affects churn prediction performance in banking industry. It is believed that those techniques have a considerably impact on recall and F1-score (Adiputra, Wanchai & Lin, 2025; Tékouabou, 2022; Adiputra, Wanchai & Lin, 2025).

Besides class imbalance handling method, hyperparameter tuning can extensively increase predictive accuracy yet it is rarely assessed across model families within a unified experimental setting (Domingos, Ojeme & Daramola, 2021; Patel *et al.*, 2023). The next question investigates the usefulness of different hyperparameter tuning methods for both ML and DL models.

Despite how well predictive models perform, they are still none-senses to business stake holders if they don't understand them. The final question address how interpretability analysis help to gain trust from entrepreneurs.

In short, this research addresses:

- How machine learning and deep learning models perform in churn prediction across multiple performance metrics, in particular banking industry.
- How predictive performance trades off with execution time for real-time deployment.
- How preprocessing and imbalance-handling techniques influence model performance.
- How different hyperparameter optimisation strategies perform across ML and DL models.
- How interpretability analysis brings more understanding to stake holders.

4. AIM & OBJECTIVES

4.1. Aim:

This paper's primary purpose is to carefully evaluate advanced predictive models for customer attrition in the banking sector, using both traditional machine learning algorithms and deep learning architectures. By conducting a methodical comparative analysis, the study aims to identify more effective solutions in terms of predictive performance, computational efficiency, and interpretability thereby offering actionable insights for improving customer retention strategies.

4.2. Objectives:

i. To review existing literature on churn prediction and identify practical gaps, particularly in comparing machine learning and deep learning algorithms in banking.
ii. To implement and evaluate machine learning and deep learning models on a common banking dataset. The performance is evaluated across multiple metrics.
iii. To analyse the trade-off between predictive performance and execution time, as well as the impact of preprocessing and class-imbalance treatments on model.
iv. To assess the effectiveness of different hyperparameter tuning methods in enhancing both machine learning and deep learning models and derive useful insights for deployment in banking sectors.
v. To explain the difficulty in predictive models to typical executives.

The table below sums up the objectives and research questions alignment.

Table 2: Objectives and Research Questions alignment

Objectives	Research questions
Review literature & identify gaps (ML vs DL in banking churn)	How do ML and DL models compare in predicting banking churn across metrics?
Implement ML & DL models and benchmark across metrics	How does predictive performance trade off with execution time in real-time deployment?
Investigate performance-execution time trade-off & preprocessing impact	How do preprocessing and imbalance-handling techniques influence model performance?
Evaluate hyperparameter optimisation strategies for ML & DL	How do different hyperparameter optimisation strategies perform across ML and DL models?
Explain predictive model to business people.	How does interpretability analysis bring more understanding to stake holders?

5. SIGNIFICANCE OF THE STUDY

This study offers banks and financial institutes an analytical recommendation on the proper predictive model application for customer retention planning. Even though predictive analytics has certain reputation in customer retention actions, it is scarce to see a head-to-head comparison of machine learning and deep learning techniques on banking specific datasets (AbdelAziz *et al*., 2025; Singh, H., *et al*., 2024). Given that absence of comparison research, banking customer retention strategies are often determined by outcome from many different settings, which may not reflect standard behaviors in financial services specific to the financial services sector.

As discussed earlier in background sections, there are some gaps between previous literature and expectation. This study seeks to narrow those gaps. First, it brings both machine learning and deep learning model under identical experimental settings, which is same datasets for the same industry. Second, it also evaluates those algorithms in term of predictive performance which are accuracy, precision, recall, F1-score, AUC and execution time. Those metrics are discussed detailly in methodology section. Execution time is often underestimated in model section studies. It is carefully evaluated in this paper, because it is essentially efficiency for real time retention systems, where milliseconds may result in wrong decision making (Singh, P., *et al*., 2025).

According to prior works, by integrating modern imbalance treatment and hyperparameter tuning can remarkably increase model sensitivity to churn (Ali, M., *et al*. 2024, Imran *et al*. 2023, and Bhuria *et al*. 2025). Hence, this study tries to apply this concept across several machine learning and deep learning in banking specific dataset. This study aims to provide a practical guidance data science teams aiming to optimize returns on analytics investments.

In banking sectors, banking practitioners are often cautious and may distrust models they cannot understand. Hence, this paper also aims to discuss explainable AI (XAI), which is said that it may help to increase interpretability. According to Li, X., *et al*. (2025), Singh, A., *et al*. (2024), and Ma *et al*. (2022), the interpretability frameworks promises that study findings are interpreted in such a way that ordinary people may grasp the underlying rationale.

This work enhances academic knowledge and industrial practice by providing a validated and replicable methodology for predicting turnover in banking. The insights will assist decision-makers in choosing models that perform well in offline evaluations while meeting the operational, regulatory, and strategic requirements of live banking environments. This puts predictive analytics as an essential facilitator of sustainable customer relationship management amid increased competition and digital change.

6. SCOPE OF THE STUDY

6.1. Inclusions:

- Analysis is limited to the provided Kaggle banking churn dataset, consisting of over 5,000 customer records with demographic, behavioural, and account-related attributes.
- Models evaluated will be confined to the specified machine learning and deep learning algorithms. The list of algorithms is described more detail in methodology section.
- Evaluation will be conducted using six metrics: accuracy, precision, recall, and F1-score, AUC and execution time. Those metrics and selection rationale are explained detailly in methodology section.
- Machine learning hyperparameter tuning will be performed using Grid Search, Random Search and Optuna. Deep learning hyperparameter will be set using prior study setups.

6.2. Exclusions:

- Real-time data streaming architectures and machine learning ops for continuous integration/continuous development (CI/CD).
- Customer churn prediction in non-banking domains.
- Other deep learning architectures such as convolutional or recurrent networks are not part of this study, because of computational constraints.
- Full hyperparameter tuning for deep learning model is excluded from this study. While machine learning models are properly tuned, deep learning models are using previous literature configurations. It may reflect a conscious trade-off between fairness, computational feasibility and scope.

7. METHODOLOGY

This study adopts a quantitative, supervised learning approach to predict customer churn in banking, enabling a controlled comparison of machine learning (ML) and deep learning (DL) methods under consistent conditions. The methodology involves data preprocessing, feature engineering, model selection, hyperparameter optimization, and evaluation, following best practices reported in recent churn prediction literature (AbdelAziz *et al*., 2025; Bhuria *et al*., 2025; Patel *et al*., 2023). This figure describe the entire methodology workflow.

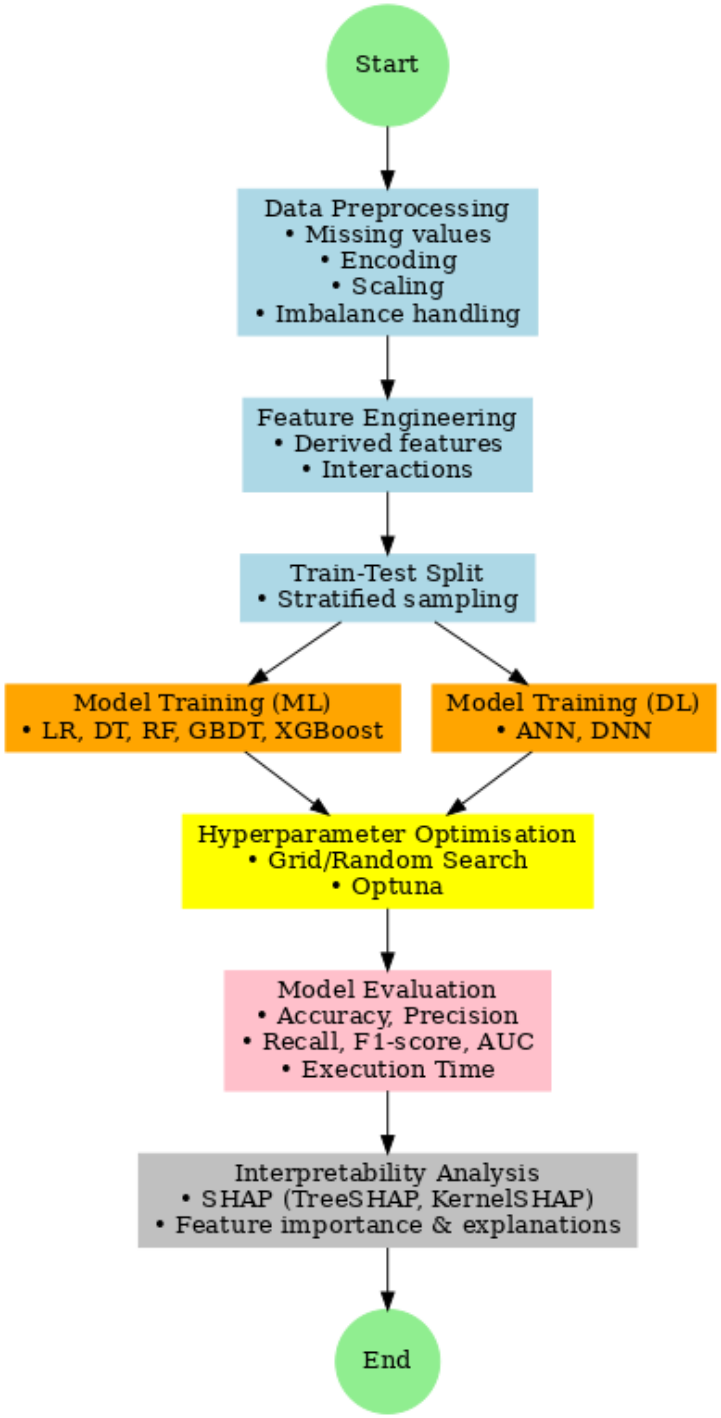

Figure 1: Research Methodology Workflow

7.1. Dataset description

This study uses the public dataset from Kaggle, which is "Churn Prediction: Unlocking Retention Secrets" (Simronw, 2023). The dataset has 5000 rows, and 5000 unique customer IDs. Thus, it is one row per customer. It does not contain historical transactions, it contains the latest related insights required to understand why customer might leave and how to retain them. There are 26 variables in this dataset, which are:

- One column is customer id
- Five categorical columns are customer demographics.
- One binary variable indicating if customers churn or not. Additional column indicating the number of months ago if churn happened.
- The rest are banking detail e.g. loan information, credit score, transaction info.

A full description of all variables, including data type, data description and their potential role in churn prediction problem is provided in Appendix A.

7.2. Data Preprocessing

Since the quality and structure of input data have a direct impact on the effectiveness and reliability of ML and DL models, data preparation is an essential step in predictive modeling (Patel *et al*., 2023; Adiputra, Wanchai & Lin, 2025). According to previous literature in churn prediction research, the data preprocessing should include missing value imputation, categorical encoding, feature scaling, class imbalance handling (Bhuria *et al*., 2025; Adiputra, Wanchai & Lin, 2025).

Missing Data Handling

Missing values data has some impact to final model training. Either it may bias the model or reduce the effective sample size. It results in poor predictive reliability (Abbas *et al*., 2023; Abubakar *et al*., 2024). Thus, it must be handled properly. There are few techniques available in prior works. First, mean imputation, median imputation or mode replaces the missing value by mean, median or mode of the feature respectively. This technique is computationally simple but highly sensitive to outliers. Using it without proper care may distort feature distribution.

Next, regression imputation and multiple imputation by chained equations (MICE) replace missing value by an approximately statically estimation. There are some advantages and disadvantages. On the good side, they leverage relationships between features which is sometime benefit to the model training. On the other hand, they require additional computational resources and often assume unnecessary predictor linearity.

Finally, k-nearest neighbour (kNN) replaces missing values by similar samples, which is believed to provide more sophisticated result. This inference consumes intensively computational resources on large dataset, hence scalability in banking churn prediction may be a full stop.

The table below summarises common missing value treatment as described above.

Table 3: Comparison of Missing Data Imputation Methods

Technique	Advantages	Limitations	Computational Cost	Citation
Mean Imputation	Simple, fast	Distorted by outliers; weak distributional fidelity	Very low	Abbas *et al.* (2023)
Median Imputation	Robust to outliers; preserves central tendency	May ignore variance structure	Low	Abbas *et al.* (2023)
Mode / Unknown Class	Suitable for categorical data; preserves absence patterns	May inflate frequency of dominant class	Low	Abubakar *et al.* (2024)
Regression Imputation	Leverages relationships between features	Assumes linearity; risk of overfitting	Medium	Murindanyi *et al.* (2025)
MICE (Multiple Imputation)	Statistically principled; handles uncertainty	Computationally expensive; complex to implement	High	Domingos, Ojeme & Daramola (2021)
kNN Imputation	Adaptive; uses neighbourhood similarity	Scales poorly with dataset size	High	Murindanyi *et al.* (2025)

This study not only focus on predictive performance but also computational efficiency. Therefore, median and mode imputation methods are applied. Numerical features (e.g., balances, transaction volumes) will be imputed using the median, which mitigates the impacts of extreme outliers compared to mean replacement. Categorical features (e.g., customer demographics) will be imputed with the mode or encoded into an explicit *"Unknown"* class. It preserves missing-data patterns as well as maintain feature interpretability. This approach ensures methodological consistency with prior banking churn studies (Abbas *et al.*, 2023; Abubakar *et al.*, 2024), while avoiding the excessive computational cost of more complex imputation frameworks.

Categorical Feature Encoding

Both ML and DL algorithms only accept numerical features, therefore it is a mandatory to transform all categorical to numerical format. There are a few encoding techniques as described in table below.

Table 4: Comparison of Encoding Methods

Encoding Method	Problem Addressed	Pros	Cons	Suitability	Citation
Label Encoding	Converts categories into integers	Simple, compact representation	• Introduces artificial ordinality • Misleads algorithms sensitive to scale	Useful for tree-based models (DT, RF, XGBoost)	Bhatia *et al.* (2023)
Target Encoding	Encodes categories by their statistical relationship to churn (e.g., mean churn rate)	• Captures category–target correlation • Reduces dimensionality	• Risk of target leakage • Requires careful cross-validation	Suitable for high-cardinality categorical features	Abbas *et al.* (2023)
One-Hot Encoding	Expands categories into binary indicator variables	• Prevents artificial ordinality • Interpretable • effective for low-cardinality data	Increases dimensionality if features have high cardinalities	Best for low-cardinality attributes (e.g., demographics) in ML/DL models	• Boozary (2025) • AbdelAziz *et al.* (2025) • Patel, S., *et al.* (2024)

The dataset includes five categorical demographic attributes, and due to low cardinalities of each features, one-hot encoding is particularly suitable method. Low cardinalities also ensure one-hot encoding computationally efficient and less prone to sparsity issues compared to high-cardinality cases. In addition, it also avoids the risk of introducing misleading ordinal relationships that could occur with label encoding, ensuring the encoded features align with the nature of the data

One-hot encoding creates a separate binary feature (0 or 1) for each category level, ensuring that ML and DL models can process categorical data without imposing artificial ordinal relationships where none exist (Boozary, 2025).

Formally, for a categorical variable C with k distinct categories $\{c_1, c_2, ..., c_k\}$, one-hot encoding maps each observation $x \in C$ to a binary vector $v \in \{0,1\}^k$ where:

$$v_i = \begin{cases} 1 \; if \; x = c_i \\ 0 \; otherwise \end{cases}$$

For example, in dataset, the gender feature has 3 values Male, Female and Other. One-hot encoding will represent these as:

Gender	**Male**	**Female**	**Other**
Male	1	0	0
Female	0	1	0
Other	0	0	1

In practice, we do not really need all three columns. Two columns suffice to derive the last columns. By removing one column, we also avoid multicollinearity.

The binary churn indicator will be retained in its original 0/1 form, following conventions in prior banking churn studies (AbdelAziz *et al.*, 2025).

Feature Scaling

Feature scaling is an important preprocessing step for all binary classification. Especially with banking datasets, numerical features (e.g. account balances, transaction volume or transaction frequency, tenor) are in various ranges. If this step is not handled properly or frequently overlooked, larger numeric scales can dominate distance-based or gradient-based algorithms. As a result, algorithms relying on distances or gradient can suffer from slow or unstable training. In the worst case, it may also lead to biased learning and unstable convergence (Ali, S., *et al*., 2025).

Hence, for those predictive algorithms which are sensitive to feature magnitude, it is a mandatory requirement to ensure uniform feature contribution. There are standardisation and normalisastion scaling methods. Normalisation scale data to a specific range which is between 0 and 1. It can distort the underlying data distribution and casually ignore meaningful patterns, especially when outliers are present. Standardisation technique rescales feature to have a mean of zero and standard deviation of one. It preserves the distribution shape while centering and scaling the data, ensuring that no single variable dominates due to its magnitude.

Standardisation is particularly important for gradient-based models such as deep neural networks (DNNs), where unscaled features can lead to inefficient convergence (Domingos, Ojeme & Daramola, 2021), and for distance-based algorithms, where differences in scale can distort similarity measures. Standardisation is mathematically defined as:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original feature value, μ is the feature mean, and σ is the feature standard deviation.

For example, if we have five features x1 to x5, their values vary from 5 to 20, this figure explains how standardisation works. The chart on the left plot five features with raw value, and the right hand side chart plots their values after standardisation.

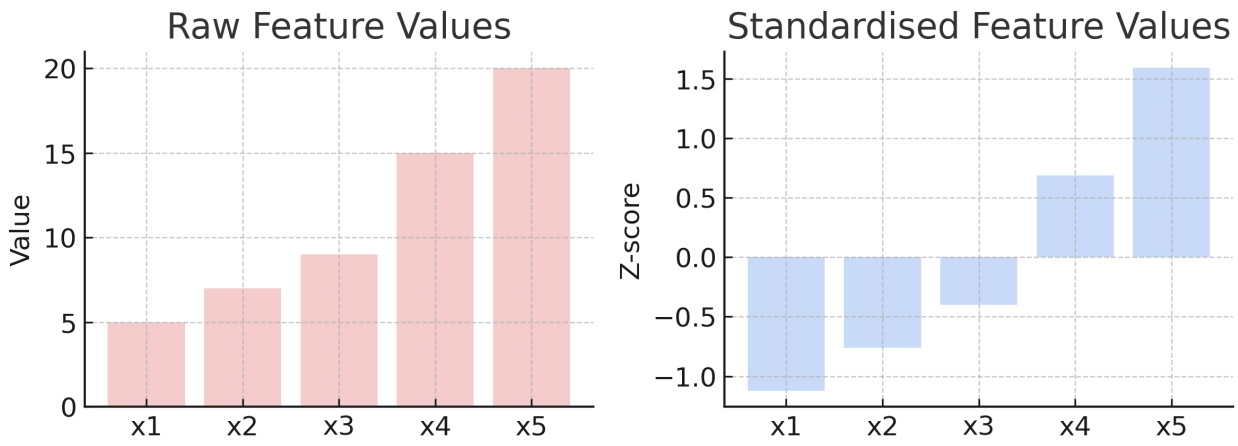

Figure 2 - Example of Standardisation scaling

Class Imbalance Treatment

For binary classification problem, particularly for churn prediction class imbalance, where one class is much of the population, is typical. It results in model biased and reduce sensitivity to true churners, which are of highest business concern. For example, if a dataset has 10000 observations, and only 2% of them are churn. By predicting all of them are non-churn, the model accuracy is 98%. Although the predictive accuracy is very high, all churn customers are ignored.

To address this, two different imbalance treatments are applied in this study as described in table below

Table 5: Comparison of Class Imbalance Treatment Techniques

Technique	Principle	Strengths	Weaknesses	Resources Cost	Best for	Citation
SMOTE	Creates synthetic minority samples by interpolating between existing samples	• Reduces overfitting from simple duplication • Improves minority representation	• May generate less realistic samples • Risk of overlapping classes	Low–Medium	Machine learning	• Adiputra, Wanchai & Lin (2025) • Tékouabou (2022)

GAN-based Oversampling	Uses a generator–discriminator framework to produce realistic minority samples	• Captures complex feature relationships • Generates highly realistic churner samples	• Computationally expensive • Risk of mode collapse in GAN training	High	Deep learning	Adiputra, Wanchai & Lin (2025)

<u>Synthetic Minority Oversampling Technique (SMOTE)</u>**:** SMOTE generates synthetic churn samples by interpolating between existing minority class instances. For each minority sample x_i, one of its k-nearest neighbours x_{nn} is selected, and a new synthetic point x_{new} is created as:

$$x_{new} = x_i + \lambda . (x_{nn} - x_i) , \lambda \sim U(0,1)$$

Where:

- x_i: a randomly chosen minority class sample
- x_{nn}: one of the k-nearest neighbours of x_i from the minority class
- λ: a random number drawn from a uniform distribution between 0 and 1 (U(0,1))
- x_{new}: the newly generated synthetic sample

This interpolation along the feature space line segment between x_i and x_{nn} avoids direct duplication while enriching the minority class distribution. Studies (Adiputra, Wanchai & Lin, 2025; Tékouabou, 2022) confirm that SMOTE improves sensitivity to churners while reducing overfitting risks associated with random oversampling.

<u>GAN-based Oversampling</u>**:** Generative Adversarial Networks (GANs) offer a more sophisticated approach by learning the underlying data distribution of the minority class. A GAN consists of two neural networks: the generator G(z), which maps random noise z ~ p(z) to synthetic samples, and the discriminator D(x), which attempts to distinguish real from synthetic samples.

Through this min–max optimisation, G learns to generate synthetic churner samples indistinguishable from real ones, thereby capturing high-order correlations and producing more realistic minority examples. GAN-based oversampling has been shown to outperform interpolation-based methods in datasets with complex feature relationships (Adiputra, Wanchai & Lin, 2025).

7.3. Feature Engineering

During exploratory data analysis process new features may be derived to capture hidden customer behavior leading to churn signals. There is a clear proof from prior literature that those derived features improving discriminative power in churn models by revealing patterns not directly observable in the raw data (Singh, H., *et al.*, 2024; Abbas *et al.*, 2023). According to previous studies and banking domain expertise, five groups of derived features are proposed (a full summary of derived features is provided in appendix B).

a. **Tenure-related features**. The customer's tenure, also known as month-on-book (MOB), is the month difference between account opening date and current data date. This feature provides a strong indicator of customer loyalty. A short tenure customers tend to be more volatile while longer tenure customers are more likely loyal (AbdelAziz *et al*., 2025). Similarly, the inactivity period, measured as the number of days since the last transaction, is another indicator for exiting, as longer inactivity period are strongly correlated with churn (Ahmed *et al*., 2023).
b. **Transaction-based features**. Metrics such as transaction frequency (number of transactions normalised by tenure), and average transaction size provide insights into how actively a customer uses their account. A balance volatility index, calculated as the absolute change in balance relative to the current balance, reflects financial instability and has been associated with higher churn tendencies.
c. **Financial health indicators**. A loan-to-income ratio will be engineered to assess leverage risk, complementing credit score information. Customers with higher ratios may face repayment stress, which increases the likelihood of discontinuing banking relationships.
d. **Service and satisfaction related**. A complaint-to-interaction ratio captures the density of dissatisfaction signals by dividing recent complaints by total customer service interactions. This is complemented by a satisfaction gap indicator, combining customer satisfaction score and complaints: low satisfaction combined with frequent complaints is a strong churn predictor, while high satisfaction and no complaints indicate stability.
e. **Contextual features** such as branch or region-level churn rates (derived from the training data) will be encoded as risk-based features, as certain regions or branches may consistently show higher attrition. A composite engagement index, integrating transaction frequency, satisfaction, and complaint ratios, will also be constructed to provide a holistic measure of customer commitment.

Completion of data preprocessing and feature engineering, the dataset will be ready and optimised for both ML and DL models. Those two steps ensure selected model evaluation exposes actual algorithm capability rather than data quality or representation.

7.4. Train–Test Split and Validation

Random sampling is usually preferred for train-test split. Each observation is assigned randomly to each train and test set. It ignores class contribution so it may produce improper split where minor class is under-sampling or even absent. Models trained on such split may underperform in churn prediction. Therefore, according to Ahmed *et al.,* (2023) and Patel *et al.,* (2023), to deal with such imbalance class problem the churn/non-churn ratio preservation is a mandatory. Instead of random sampling, stratified sampling should be employed to split training (80%) and testing (20%) subsets. Stratification guarantees that both subsets reflect the same class distribution as the original dataset, which is critical for imbalanced problems such as churn prediction.

Moreover, as another recommendation by Marmion *et al.*, (2012), k-fold cross validation will be applied during model training to limit overfitting and reduce variance in final prediction. The train dataset is equally split further down by k folds, and the model is trained k times. Each time, it leaves one fold out for validation and uses remaining folds for training. The final performance estimation is computed as the average across all k runs. Formally, if D is the training dataset and F_1, F_2, ..., F_k are the folds, the model is trained k times on all folds except F_i and validated on F_i. This approach is also useful of limited data.

For example, if we choose k equal to 5, this figure demonstrate how k-fold validation works.

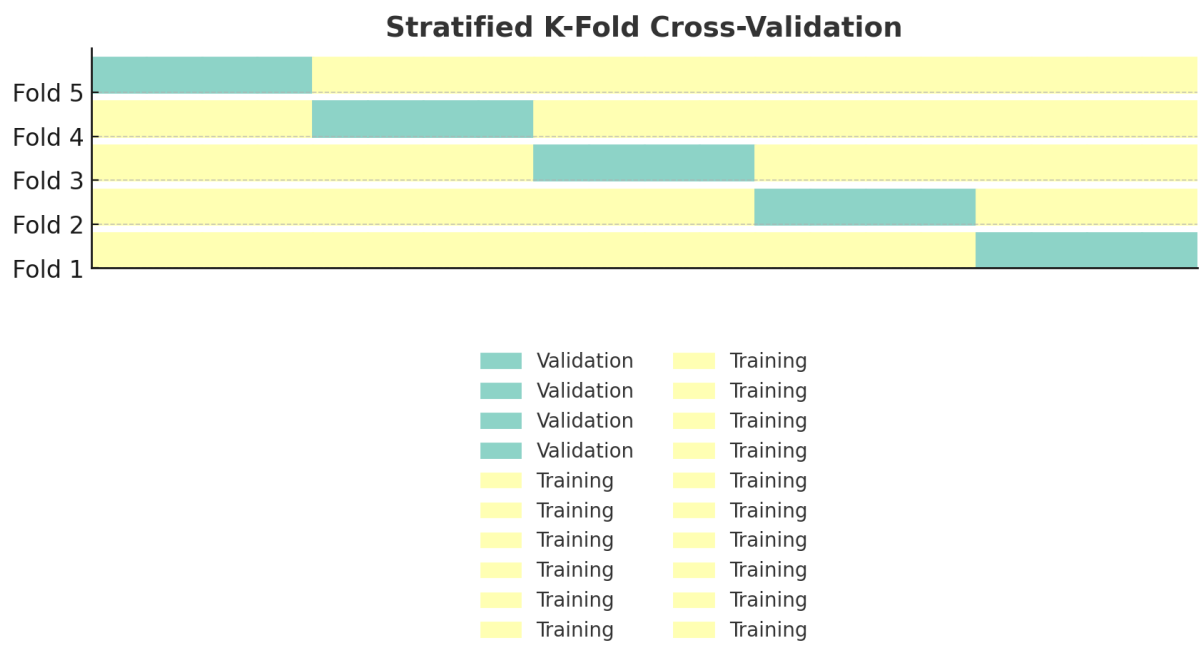

Figure 3 - Demonstration of stratified k-Fold Cross-Validation

7.5. Algorithms Considered

There are two objectives behind the model selection in this study:

- Ensuring covering of various linear classifiers, decision-tree-based ensembles, and non-linear representation algorithms.
- Enabling a balanced comparison between ML and DL approaches using consistent preprocessing and evaluation pipelines.

This ensures the experimental outcomes are both statistically rigorous and operationally meaningful in a banking context.

Comparative Perspective (ML vs. DL in the four-core metrics)

Evaluation metrics will be discussed furtherly in the evaluation metric part in this section. Evidence from the reviewed literature consistently reveals complementary strengths between ML and DL in banking churn prediction:

Table 6: Model performance comparison

Metric	Best-performing ML models	Best-performing DL models	Observed trends in banking churn prediction
Accuracy	XGBoost, Random Forest	Deep Neural Network (tuned)	• Ensembles often lead due to bias–variance optimisation • DNN can match with careful tuning.
Precision	Random Forest, Logistic Regression	Artificial Neural Network	• ML ensembles typically produce fewer false positives • ANN can offer competitive precision with feature scaling.
Recall	XGBoost (with rebalancing)	Deep Neural Network	• DNN often excels in recall, especially with dropout and class weighting • Important for catching true churners.
F1-score	XGBoost	Deep Neural Network	• Context-dependent: boosting models lead on clean, separable data • DNN can surpass with complex feature interactions.
Execution time	Logistic Regression, Decision Tree	Artificial Neural Network	• Simpler models faster • ANN reasonable latency

This comparative evidence motivates the inclusion of models that mutually span the metric range: ensembles (RF, XGBoost) for accuracy and precision, and DL models (DNN, ANN) for recall and balanced F1 performance. Logistic Regression and Decision Trees serve as low-latency baselines and interpretable respectively, enabling a full assessment of performance versus execution time.

Machine Learning Models

- **Logistic Regression (LR).** A linear probabilistic classifier that models the log-odds of churn as a function of customer attributes. LR remains a robust baseline in banking churn research due to its transparency, calibration friendliness, and low computational cost (Abbas *et al.*, 2023; AbdelAziz *et al.*, 2025; Patel *et al.*, 2024).
- **Decision Tree (DT).** A hierarchical partitioning of the feature space that yields human-readable rules and very fast inference. While single trees often trail ensembles in raw accuracy, their latency and interpretability are attractive for real-time scoring and compliance-centric environments (Bhuria *et al.,* 2025; Tékouabou, 2022).
- **Random Forest (RF).** An ensemble of decorrelated trees trained via bagging and random feature selection. Therefore, RF typically improves generalisation and stability relative to a single DT. On the other hand, RF requires way more computational resources and so longer execution time than DT. Since this study also values execution time and computational resources, DT and RF are evaluated head-to-head to consider the trade-off between those execution time and predictive performance (Bhuria *et al.*, 2025; Singh, P., *et al.*, 2025).
- **XGBoost.** Sequential ensembles that correct residual errors of prior learners and incorporate regularisation; they have repeatedly achieved state-of-the-art results on structured banking data, especially when hyperparameters are tuned systematically (Patel *et al.,* 2023; Ali, M., *et al.,* 2024; AbdelAziz *et al.,* 2025; Li, X., *et al.,* 2025). In previous literature as summarized in table above, XGBoost beats all other three machine learning model in term of accuracy, recall and F1-score. Like RF, this model also requires more computational cost. It is also good to consider the trade-off between predictive performance and computational cost.

Deep Learning Models

- **Artificial Neural Network (ANN):** Feed-forward multi-layer perceptions modelling non-linear relationships between demographic, behavioural, and transactional features and churn likelihood. Typical configurations in banking churn tasks include 1–2 hidden layers with ReLU activation, dropout for regularisation, and a sigmoid output neuron for binary classification (AbdelAziz *et al*., 2025; Adamu *et al*., 2025). Like LR in machine learning model, this algorithm acts as baseline in churn prediction for deep learning.
- **Deep Neural Network (DNN):** Extends the ANN by incorporating additional hidden layers (e.g., 4–6) to capture higher-order feature interactions in high-dimensional data. Common designs employ dense layers with ReLU activation, batch normalisation to stabilise training, dropout to mitigate overfitting, and a sigmoid output layer (Singh, H., *et al*., 2024; Basit *et al*., 2024). Because of complex architecture, DNN wins most predictive performance, yet it consumes more resources. In term of trade-off consideration, both DNN and ANN are put on the table for consideration.

7.6. Hyperparameter Optimisation

In all predictive algorithms, there are two types of parameters. First, it is model parameters such as regression coefficients or neural network weights learning and updating during the training using data. The other is model hyperparameters controlling learning process itself rather than learning from data. Some examples include the regularisation strength (e.g., L1 or L2 penalty) for logistic regression or maximum tree depth, minimum samples per split for decision tree. They are set prior to training and directly influence model capacity, convergence behaviour, and generalisation ability (Patel *et al*., 2023; Abubakar *et al*., 2024).

Machine Learning Models

Previous literatures believe the importance of hyperparameter tuning for maximizing the predictive performance of both ML and DL models. In contrast, deficient configurations can lead to either underfitting or overfitting. This study applies comparative approach using grid search, random search, and Optuna.

- **Grid Search:** This method brute forces all possible combinations of predefined hyperparameters. By covering all possibilities, it can determine the best hyperparameter combination. Once the parameter space grows, the good application becomes computational burden (Abubakar *et al*., 2024). For example, tuning three parameters with ten candidate values each, formally ten power of three, already requires 1,000 model fits, making grid search infeasible for complex DL architectures.
- **Random Search:** Instead of testing all combinations, random search randomly selects some of hyperparameter configurations. Obviously, it performs faster than grid search and reduces computational resources. Nonetheless, it may still spend trials on nonpotential regions of parameter spaces and lacks adaptive exploration (Abubakar *et al*., 2024) or it may miss the best combination.
- **Optuna:** A modern hyperparameter optimization framework that improves on both methods by employing Bayesian optimization and early pruning strategies (Patel *et al*., 2023). Bayesian optimization uses probabilistic models to guide the search towards potential regions of the parameter space. On the other hand, pruning stops disappointing trials early, reallocate resources to more potential candidates. This combination returns greater efficiency in exploring complex search spaces, particularly for deep neural networks where training iterations are computationally expensive.

Table 7: Comparison of Hyperparameter Optimisation Approaches for Machine Learning Models

Approach	Description	Strengths	Weaknesses	Relevant Studies
Grid Search	Exhaustively evaluates all parameter combinations within a predefined grid.	Simple to implement; guarantees finding best model within the grid.	Computationally expensive; infeasible for large search spaces.	Abubakar *et al*. (2024)

Random Search	Samples parameter values randomly from defined distributions.	More efficient than grid search in high-dimensional spaces; can reach near-optimal results quickly.	May miss optimal combinations if not sampled; results can vary by run.	Patel *et al*. (2023); Abubakar *et al*. (2024)
Optuna	Uses Bayesian optimisation and pruning to guide search toward promising regions.	Highly efficient in large, complex spaces; adaptive search; early stopping of poor trials saves computation.	More complex implementation; requires tuning of optimisation process itself.	Patel *et al*. (2023)

In conclusion, logically Optuna beats Grid Search and Random Search not only in term of computational resources but also achieving higher chance of finding near optimal. By applying all three approaches and comparing each approach's peak performance with base models and them together, the study seeks the solid proof for the best tuning platform.

Deep Learning Models

In contrast, DL models in this study will not undergo full hyperparameter tuning process. Due to deep learning complex architectures, hyperparameter tuning requires extremely high computational resources. Their hyperparameters such as number of hidden layers, activation functions, dropout, batch size, learning rate will be set using configurations established in prior churn prediction studies (Amal *et al*., 2023; Ahmed *et al*., 2023; Zhang *et al*., 2024). This reflects two concerns:

- Because the study dataset is quite small (5000 observations) and it may limit the extensive hyperparameter tuning benefit. Any overly complex tuning may risk to overfitting.
- Full hyperparameter tuning is computationally expensive and beyond the practical scope of this study, as the scop

By adopting configurations from previous literature, DL models are not left untuned but are setup using empirically validated settings. This ensures methodological fairness, ML models are fully tuned with Grid Search, Random Search, and Optuna, while DL models are reasonably configured to reflect their potential under realistic study constraints.

7.7. Evaluation Metrics

Besides four widely used metrics in the binary classification problem (Adiputra, Wanchai & Lin, 2025; Bhuria *et al*., 2025), the study will also consider two extra metrics, which are AUC (Ahmed *et al*., 2023; Patel *et al*., 2023) and execution time (Bhuria *et al*., 2025; Domingos, Ojeme & Daramola, 2021).

This below figure denotes some formulas below (Tharwat 2018):

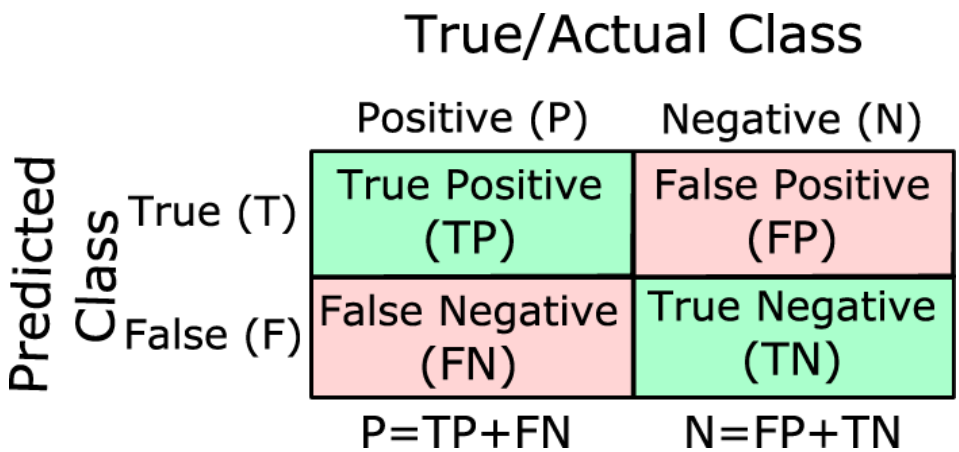

Figure 4: Confusion matrix

- **TP** is true positive; a churn customer is predicted to churn correctly.
- **TN** is true negative; a non-churn customer is predicted to be a non-churn customer correctly.
- **FP** is false positive; a non-churn customer is predicted to churn incorrectly.
- **FN** is false negative; a churn customer is predicted to be a non-churn incorrectly.

a. **Accuracy** is a metric that quantifies how often a model's predictions are correct. The term "correct" here refers to accurate classification of both churn and non-churn classes. It is a good starting point for evaluating model performance because it is simple to understand and calculate, making it a good initial measure of how well the model is performing.

$$Accuracy = \frac{Number\ of\ correct\ prediction}{Total\ number\ of\ predictions} = \frac{TP + TN}{TP + TN + FP + FN}$$

However, the metric itself is sometimes interpreted wrongly especially for high imbalance class prediction. By predicting all records are majority class, the model achieves high accuracy score, but it fails to reflex the real situation. Therefore, additional metrics are used alongside the evaluation to make it clearer.

b. **Precision** is a metric that measures the accuracy of positive predictions produced by a model. In another word, precision is the proportion of correctly predicted positives out of all instances predicted as positive. This metric indicates the proportion of actual positives among those predicted as positive by the model. Precision is used to evaluate the reliability

of a model's positive predictions and can be particularly useful in situations where false positives may have a significant impact. For example, in customer retention scenarios, banks may offer promotions; a higher number of false positives may result in increased costs for the bank.

$$Precision = \frac{TP}{TP + FP}$$

c. **Recall,** also known as true positive rate (TPR) or sensitivity, quantifies a model's ability to identify relevant instances (true positives) within a dataset. It represents the proportion of actual positive cases correctly detected by the model, indicating the fraction of positives found among all actual positives.

$$Recall = \frac{TP}{TP + FN}$$

Recall is given particular emphasis in banking churn contexts, as failing to identify an at-risk customer may result in direct revenue loss (Tékouabou, 2022; Adiputra, Wanchai & Lin, 2025). While precision focuses on the accuracy of positive predictions, recall focuses on identifying all positives. A high recall can sometimes come at the cost of lower precision (more false positives), and vice versa. The relationship between precision and recall is often described as a trade-off.

d. **F1-score** provides a balanced measure of a model's performance in classification tasks, particularly useful when dealing with imbalanced datasets. It is the harmonic mean of precision and recall, offering a single value that considers both false positives and false negatives:

$$F1 = 2\,x\,\frac{Precision\ x\ Recall}{Precision + Recall}$$

The F1-score provides a single, comparable metric for evaluating and comparing the performance of different machine learning models, especially when precision and recall are both important. A high F1-score (close to 1) signifies that the model has outstanding precision and recall, meaning it is effective at both identifying positive cases and minimizing incorrect positive predictions. A low F1-score (close to 0) indicates that the model is either not identifying enough positive cases (low recall) or is making too many incorrect positive predictions (low precision), or both. In banking, either failing to retain a valuable customer or offering unnecessary incentives entail substantial cost.

e. **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**: a threshold-independent measure representing the probability that the model ranks a randomly chosen churner higher than a randomly chosen non-churner. The ROC curve plots the true positive rate (recall) against the false positive rate at various classification thresholds, and the AUC quantifies the overall separability. In general classification problems, higher AUC values indicate better discrimination between classes. In churn prediction, AUC is especially valuable because it evaluates performance across all possible thresholds, helping

practitioners select the most appropriate operational cut-off point for intervention strategies (Ahmed *et al*., 2023; Patel *et al*., 2023).

f. **Execution Time** measured in seconds or milliseconds per inference or training epoch, depending on model type. It captures computational efficiency during both training and prediction phases. Built-in profiling tools provide precise measurement of both CPU and GPU operations. Execution time will be calculated by time differences between wall-clock time at beginning and end of each training run or predicting. To be fair, time both training and predicting time execution will be normalized to per-epoch or per-sample

$$Train_{time} = \frac{endtime - startime}{training\ set} \qquad Pred_{time} = \frac{endtime - startime}{predicting\ set}$$

Table 8: Runtime Benchmarking Protocol for Execution Time

Aspect	Description
Measurement primitives	• For ML models: use high-resolution wall-clock timers (e.g., Python time.perf_counter()). • For DL models running on GPU: use CUDA synchronizers (torch.cuda.synchronize()) to account for asynchronous kernel execution.
Warm-up & repetitions	• Perform 1–3 warm-up runs to populate caches and JIT compile graphs. • Execute 10 independent timed runs. • Report median (p50) and optionally p90 latency. • Outliers are logged and reported separately.
What is timed	• Training: fit() call (ML) or forward+backward passes (DL). • Inference: predict() (ML) or forward pass (DL). • Include preprocessing/transform steps required inside pipeline.
What is not timed	• Disk I/O, dataset loading, or caching operations. • One-off model/graph compilation (outside warm-up). • Hyperparameter search orchestration overhead.
Hardware specification	• Record CPU/GPU model, RAM, and storage configuration. • Note whether execution uses single-thread, multi-thread, or GPU acceleration.
Batch size & concurrency	• Report latency for single-sample (online scoring) and batch (mini-batch) inference. • Vary concurrency levels (1, 4, 8 workers) to simulate deployment load.
Reporting	• Present results in milliseconds per inference (prediction) and seconds per epoch (training). • Provide confidence intervals (95%) to demonstrate stability of results.

While more complex architectures (e.g., deep neural networks) may deliver marginal returns in predictive metrics, they often incur substantial delays in prediction speed and higher hardware costs. In churn prediction, where timeliness can influence retention intervention strategies, execution time becomes a crucial operational metric.

- Fast models (e.g., Logistic Regression, Decision Tree) can be deployed easily and score lower in complexity but may sacrifice some accuracy.
- The inclusion of execution time allows decision-makers to balance predictive power with operational feasibility.

g. **Evaluation Metrics and Statistical Significance**: a model with higher raw metrics (accuracy, precision, recall, F1, execution time) is an efficient requirements to claim this model outperform other models. Statistical significance testing helps to ensure the evaluation is sufficiently rigorous to conclude that higher performance does not happen by chance or by sampling. The table below describes how each five above evaluation metrics are statistically tested.

Table 9: Evaluation Metrics and Statistical Significance using Cross Validation

Metric	Definition	How It is Compared (with CV)	Ensuring Robustness
Accuracy	Ratio of correctly classified cases over total cases.	• Compute fold-level accuracy in repeated stratified k-fold CV • Use the distribution of scores across folds to compare models.	Apply paired tests (paired *t*-test or Wilcoxon signed-rank) or bootstrap 95% confidence intervals to assess statistical significance (Marmion *et al*., 2012).
Precision	Fraction of predicted churners that are actual churners.	• Compute fold-level precision in repeated stratified k-fold CV • Use the distribution of scores across folds to compare models.	Same significance logic as accuracy: paired tests or bootstrap 95% CIs ensure results are not due to chance.
Recall	Fraction of actual churners correctly identified (especially important in churn contex.	• Compute fold-level recall in repeated stratified k-fold CV • Use the distribution of scores across folds to compare models.	Same significance logic as accuracy: paired tests or bootstrap 95% CIs validate recall improvements across models.

F1-score	Harmonic mean of precision and recall.	• Compute fold-level F1 values in repeated stratified k-fold CV • Use the distribution of scores across folds to compare models.	Same significance logic as accuracy: paired tests or bootstrap 95% CIs ensure F1 differences are robust.
Execution Time	Time for training and inference (as per benchmarking protocol).	• Measure training/inference time within each CV fold (median and p90 latency).	Apply non-parametric tests (e.g., Kruskal–Wallis) if execution time distributions are skewed.

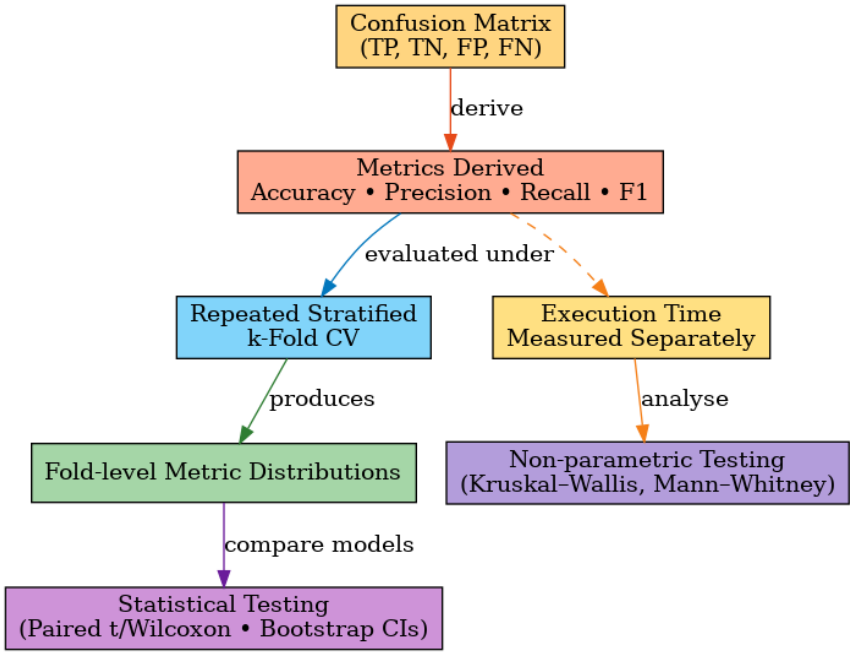

Figure 5: Workflow of Statistical Significance Test

All classification metrics (accuracy, precision, recall, and F1) are fractions derived from the same confusion matrix and evaluated identically across repeated stratified k-fold CV. By treating the fold-level values as samples from the performance distribution, it is methodologically consistent to apply a unified statistical framework either paired tests or bootstrap confidence intervals to compare models. Although these metrics are fraction-based and could in principle be tested with a binomial distribution. This overlooks the more relevant question of whether performance differences are consistent across folds. Fold-level tests or confidence intervals are therefore preferred, as they capture both model stability and generalisation variance. Execution time is treated separately because it is a skewed, continuous variable, for which non-parametric tests such as Kruskal–Wallis or Mann–Whitney is more appropriate.

7.8. Interpretability Analysis

As described in objective section, beside quantitative evaluation metrics, model interpretability is another key component of the analysis phase. In the banking industry, the ability to explain predictions is critical not only for business stake holders such as compliance, risk management, and but also is crucial for gaining executives trust (Boozary, 2025). Well performance model, which fails to explain their predictions, are inappropriate for customer-facing decision systems deployment. Because those decision may directly result in poor customer retention strategies.

This research will employ SHapley Additive exPlanations (SHAP) to provide a unified, model-agnostic approach to interpreting feature contributions. SHAP is built in cooperative game theory. Each feature contribution is computed as the average marginal contribution across all possible feature subsets. There are two main advantages of this approach. First, it generates overall explanations, which highlight features having most impact in churn prediction across entire dataset. Second, it also explains in detail why a specific customer is classified as likely to churn (AbdelAziz *et al*., 2025).

For tree-based models (e.g., Random Forest, Gradient Boosted Trees, XGBoost), TreeSHAP will be used to optimize computational efficiency. For deep learning models, KernelSHAP will approximate Shapley values via sampling, ensuring interpretability remains feasible despite higher model complexity (Domingos, Ojeme & Daramola, 2021).

The interpretability results will serve three purposes:

a. **Feature importance validation** involves cross-checking whether the highly weighted features correspond with domain expertise and existing banking literature.
b. **Model debugging** contains identifying whether specific features are introducing bias or creating false correlations.
c. **Actionable insights**: to guide retention strategies by revealing key behavioral or demographic drivers of churn.

By integrating SHAP analysis into the evaluation pipeline, the study will bridge the gap between predictive accuracy and actionable decision-making, ensuring that model outputs are not only correct but also transparent and trustworthy.

8. REQUIRED RESOURCES

- **Hardware:** a cloud-based environment with GPU acceleration. Minimum specification is 16 GB RAM, SSD storage to enable fast data processing and NVIDIA CUDA-enabled GPU for deep learning model training.
- **Software:** Python 3.10 with machine learning libraries such as Scikit-learn, TensorFlow, PyTorch, profiling tools, and interpretability frameworks (SHAP, LIME). All analytics are done on IDE Jupyter Notebook.
- **Dataset:** Kaggle Churn The prediction dataset (Simronw, 2023) described earlier in the scope of the study and methodology section.
- **Reference management tool:** Mendeley.

9. RESEARCH PLAN

ADVANCED PREDICTIVE MODELLING FOR CUSTOMER CHURN IN BANKING:
A COMPARATIVE ANALYSIS OF MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

Duc Le
Master of Machine Learning and Deep Learning
Research start date 3/8/25

Legend: On track | Goal

Milestone description	Category	Progress	Date start	Duration (in week)
Literature review	On Track	100%	3/8/25	4
Problem Refinement & Dataset Preparation	On Track	40%	31/8/25	2
Model Development	Goal	0%	14/9/25	4
Model Evaluation	Goal	0%	12/10/25	3
Interpretability Analysis	Goal	0%	2/11/25	3
Resource & Constraint Review	Goal	0%	23/11/25	1
Documentation & Report Writing	Goal	0%	30/11/25	3

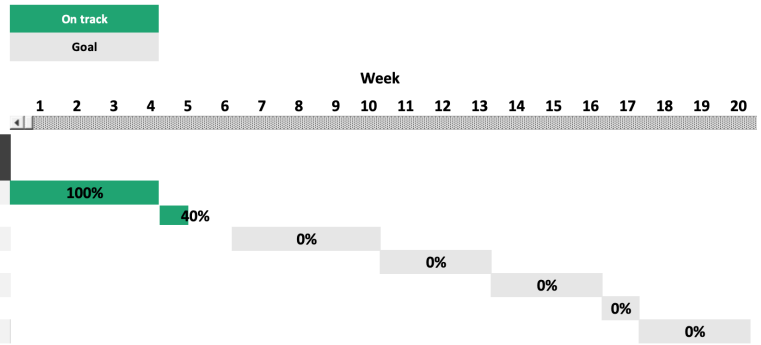

Figure 6 - Research plan

Table 10: Table of Detail Research Plan

Phase	Duration	Activities	Risk	Contingency Plans	Percentage Completed
Literature Review	Weeks 1–4	Systematic review of scholarly work on churn prediction, imbalanced learning (SMOTE, GAN), and interpretability methods (SHAP, LIME).	Limited access to high-quality articles or databases.	Use alternative databases (Google Scholar, ResearchGate) and request institutional library support.	100%
Problem Refinement & Dataset Preparation	Weeks 5–6	Define research problem, collect/preprocess datasets, perform exploratory data analysis, apply preliminary oversampling.	Dataset imbalance persists or poor data quality.	Generate synthetic data, adjust preprocessing techniques, and refine feature selection.	40% done (done the research problem definition)

Model Development	Weeks 7–10	Implement baseline ML models (LR, DT, RF, GBM) and advanced DL models; integrate GAN-based oversampling.	High computational load causing delays.	Utilise cloud-based GPUs or scale down model complexity temporarily.	0%
Model Evaluation	Weeks 11–13	Evaluate models with metrics (precision, recall, F1, AUC, execution time); compare oversampling impacts.	Conflicting metric outcomes may emerge.	Incorporate multi-metric analysis and consult domain experts to refine evaluation criteria.	0%
Interpretability Analysis	Weeks 14–16	Apply SHAP and LIME to evaluate transparency and interpret model decisions.	Interpretability results are inconclusive or difficult to explain.	Use alternative interpretability tools (e.g., PDP, ALE) and provide qualitative explanation.	0%
Resource & Constraint Review	Week 17	Review computational requirements, identify bottlenecks, refine experimental design.	Unexpected hardware/software limitations.	Switch to institutional HPC resources or optimise code efficiency.	0%
Documentation & Report Writing	Weeks 18–20	Compile findings, structure report, discuss implications, limitations, and future work.	Time constraints leading to incomplete documentation.	Begin drafting earlier in parallel with experiments and allocate extra time for revision.	0%

10.LIMITATIONS

This study acknowledges several methodological limitations, each of which has been considered carefully yet excluded from scope due to practical constraints.

a. Hyperparameter tuning for deep learning models

   While machine learning models (LR, DT, RF, XGBoost) will undergo systematic hyperparameter optimisation using Grid Search, Random Search, and Optuna, deep learning models (ANN, DNN) will not be fully tuned. This design decision rests on three considerations:

   - The dataset is relatively small (5,000 observations), which limits the marginal benefit of extensive hyperparameter search; overly complex tuning risks overfitting.
   - Full hyperparameter tuning for DL is computationally expensive, requiring hardware and time resources beyond the practical scope of this research.
   - Prior churn prediction studies provide empirically validated DL configurations that can be reasonably applied without exhaustive search (Amal *et al*., 2023; Ahmed *et al*., 2023; Zhang *et al*., 2024).

   Consequently, DL models are included as fair, literature-informed benchmarks rather than fully optimised candidates.

b. Scope of architectures

   Only baseline ANN and DNN are considered. More advanced deep learning architectures (e.g., CNNs, LSTMs, hybrid networks) are excluded, again due to computational constraints and dataset size limitations. Their exclusion narrows generalizability but ensures feasibility within a master's-level research.

c. Dataset coverage

   The dataset is cross-sectional (5,000 customers, one row per customer) and lacks historical transactional sequences. As such, the study cannot evaluate time-series models or capture longitudinal churn behaviours. This limits the temporal dimension of churn prediction but reflects the constraints of the chosen data source.

d. Generalisability

   Results will be specific to the Kaggle dataset used. While the methodology is transferable, absolute performance levels may differ in proprietary banking datasets with richer features or larger sample sizes.

These limitations are explicitly recognised as design trade-offs: the study prioritises methodological clarity, computational feasibility, and fairness over exhaustive model exploration. Future research should extend hyperparameter optimisation to deep learning architectures,

incorporate temporal datasets, and evaluate a wider range of neural models under larger-scale conditions.

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APPENDIX

Appendix A: Dataset Structure and Description

Variable	Type	Description	Notes on Use in Churn Prediction
Customer_ID	Categorical (ID)	Unique identifier for each customer	Used only as key, not for modelling
Age	Numeric	Customer's age	May capture churn patterns across age groups
Gender	Categorical	Male, Female, Other	Encoded to examine demographic churn effects
Account_Type	Categorical	Checking, Savings, Investment	Different account types may have different attrition risk
Account_Balance	Numeric	Current balance in account	Low balances may indicate disengagement
Transaction_Date	Date	Date of last transaction	Used to derive inactivity period
Transaction_Amount	Numeric	Amount of last transaction	Proxy for engagement level
Transaction_Type	Categorical	Deposit, Withdrawal, etc.	Encoded to reflect recent financial behaviour
Branch	Categorical	Branch location	May reveal regional churn hotspots
Loan_Amount	Numeric	Value of any outstanding loan	High loans relative to income may correlate with churn
Credit_Score	Numeric	Financial health indicator	Lower scores often correlate with churn
Is_Employed	Boolean	Employment status	Employment instability may increase churn likelihood
Annual_Income	Numeric	Reported yearly income	Used to derive loan-to-income ratio

Marital_Status	Categorical	Single, Married, etc.	Marital status may affect retention behaviours
Region	Categorical	Customer's geographical region	Can capture regional churn patterns
Account_Open_Date	Date	Date account was opened	Used to derive customer tenure
Last_Transaction_Date	Date	Date of most recent transaction	Used to compute inactivity features
Number_of_Transactions	Numeric	Transactions in past year	Used for transaction frequency feature
Account_Activity_Trend	Categorical	Increasing, Decreasing, Stable	Encodes behavioural trajectory
Customer_Service_Interactions	Numeric	Number of customer service interactions	High volume may indicate dissatisfaction
Recent_Complaints	Numeric	Count of complaints filed	Strong negative churn predictor
Change_in_Account_Balance	Numeric	Net change in balance over time	High fluctuations may indicate instability
Customer_Satisfaction_Score	Ordinal (1–5)	Self-reported satisfaction	Lower scores correlate with higher churn
Churn_Label	Binary	1 = churned, 0 = not churned	Target variable
Churn_Timeframe	Numeric	Months since churn (if churned)	Temporal churn intensity measure

These "notes on use in churn prediction" are solely based on previous studies and banking domain expertise. They will be validated during EDA.

Appendix B: Proposed Engineered Features

Feature	Formula / Definition	Churn Rationale	Supporting Literature
Customer Tenure	Current Date – Account_Open_Date (months)	Short-tenure customers are more likely to churn; tenure is a proxy for loyalty.	AbdelAziz *et al*. (2025)
Inactivity Period	Current Date – Last_Transaction_Date (days)	Longer inactivity indicates disengagement and higher churn risk.	Ahmed *et al*. (2023)
Transaction Frequency	Number_of_Transactions ÷ Tenure (months)	Declining activity frequency is a known churn signal.	Singh, H., *et al*. (2024)
Average Transaction Size	Total Transaction Amount ÷ Number_of_Transactions	Smaller or irregular transactions may indicate lower engagement.	Abbas *et al*. (2023)
Balance Volatility Index	Change_in_Account_Balance	÷ Account_Balance	
Loan-to-Income Ratio	Loan_Amount ÷ Annual_Income	High leverage signals repayment stress and financial instability.	Ali, S., *et al*. (2025)
Complaint-to-Interaction Ratio	Recent_Complaints ÷ (Customer_Service_Interactions + 1)	High complaint intensity indicates dissatisfaction.	Tékouabou (2022)
Satisfaction Gap Indicator	Composite of Customer_Satisfaction_Score and Recent_Complaints	Low score + high complaints = strong churn predictor.	Singh, A., *et al*. (2024)
Region/Branch Churn Rate	Derived churn rate per region/branch from training data	Captures geographical/regional differences in churn behaviour.	Basit *et al*. (2024)
Engagement Index	Weighted composite of activity frequency, satisfaction score, and complaint ratio	Holistic measure of customer commitment and loyalty.	Abbas *et al*. (2023)

These proposed derived features are solely based on previous studies and banking domain expertise. They will be validated during EDA.